INCH-POUND

MIL-C-24643/16D
22 November 1994
SUPERSEDING
MIL-C-24643/16C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSTSGU
(INCLUDING VARIATION LSTSGA)

This specification is approved for the use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

Sizes 3 to 100 inclusive, extruded silicone rubber insulation 1/

First - Copper conductor, uncoated, (see table I for size)
Second - Extruded Silicone rubber insulation (see table I for diameter).
Third - Glass braid. 80% minimum coverage.
Fourth - Braid covering (see table I for diameter). Standard identification code applied by method 1 or method 3.

Sizes 23 to 400 inclusive, silicone rubber-glass tape insulation 1/

First - Copper conductor, uncoated, (see table I for size)
Second - Silicone rubber - glass tape insulation (see table I for diameter). Letter identification code applied by method 5.
* Third - The three conductors shall be cabled together with a lay not greater then 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Fourth - A binder.

1/ Size 23 through 100 may be glass tape or extruded silicone rubber insulation.

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TABLE I. Details.

Military Part no. M24643/16 1/	Size 2/	Conductors size		Type's overall diameter (inches)				Diameter over outer insulation	Diameter over insulation	Diameter over braid	Jacket thickness min	Conductor resistance	Insulation resistance	Cold bending mandrel diameter	Accelerated service loading
				LSTSGU		LSTSGA									
		Navy std	AWG	min	max	max	max	Tape nom. (inch)	min (inch)	max (inch)	avg (inch)	(ohms)	(meg-ohms)	(max) (inches)	ampere
-01UN	3		16 Cl B		0.411		0.461	----	0.096	0.130	0.030	4.3	500	----	----
-02UN	4		14 Cl B		.449		.499	----	.112	.143	.030	2.68	500	----	----
-03UN	9		10 Cl B		.575		.625	----	.154	.187	.040	1.08	500	----	----
-04UN	14		9 Cl B		.718		.768	----	.230	.262	.040	.859	500	----	90
-05UN	23		7 Cl B		.812		.862	0.274	.271	.310	.050	.543	500	----	115
-055UN	30	30 (19)			.852		.902	0.282	.285	.335	.050	.358	225	10	140
-06UN	50		3 Cl C		.969		1.019	.334	.335	.380	.050	.21	200	12	185
-07UN	75		1 Cl C		1.134		1.185	.407	.407	.440	.050	.134	175	14	250
-08UN	100		0 Cl D		1.266		1.316	.453	.453	.500	.060	.106	160	16	295
-085UN	125	125 (61)			1.408		1.458	.507	----	----	.060	.0888	150	17	340
-09UN	150		000 Cl D		1.515		1.565	.557	----	----	.060	.0674	135	19	400
-10UN	200		0000 Cl D		1.669		1.719	.634	----	----	.060	.053	125	21	460
-105UN	250	250 (61)			1.794		1.844	.697	----	----	.060	.0444	120	22	535
-11UN	300		300 MCM Cl D		1.957		2.007	.748	----	----	.075	.0377	110	24	590
-01105un	350	350 (91)			2.073		2.123	.802	----	----	.075	.0316	105	25	665
-12UN	400	400 (127)			2.203		2.253	.862	----	----	.075	.0273	100	27	750

1/ Unarmored. When armored option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSTSGU and LSTSGA

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Construction (watertight with circuit integrity) (continued)

Construction variant type LSTSGU

Sizes 3 to 100 inclusive, extruded silicone rubber insulation 1/

* Fifth - The three conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly
Sixth - Binder or combination binder and barrier.
Seventh - Cross-linked polyolefin jacket. (see table I for thickness).

Sizes 23 to 400 inclusive, silicone rubber-glass tape insulation 1/

Fifth - Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant type LSTSGA

Fifth - Cross-linked polyolefin jacket. (see table I for thickness).
Sixth - Braided metal armor

1/ Size 23 through 100 may be glass tape or extruded silicone rubber insulation.

EXAMINATION AND TESTS:	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	
Sizes 3 through 9	3000
Sizes 14 and larger	5000
Insulation resistance - megohms/1000 feet, minimum	(see table I)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
* Abrasion resistance - sizes 3 through 100 (extruded silicone rubber) scrapes, minimum	250
* Crack resistance - sizes 3 through 100 (extruded silicone rubber)	No damage
Group B:	
Cross-linked proof test (percent, maximum	
Insulation	50
Jacket (When tested at 200°C)	50
Gas flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - see table I for mandrel diameter)	No damage

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EXAMINATION AND TESTS (continued)

	Requirements
Physicals (Unaged)	
Insulation (extruded)	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (conductor, method 1 only) - cycles, minimum (extruded silicon rubber only)	25
* Permanence of printing (jacket) - cycles, minimum	125
Cable sealant removability	No failure
Armor (type LSTSGA only) - conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Cold working (minus 20 ± 2°C)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Accelerated service - sizes 14 and larger (see table I for load current)	No failure
Aging and compatibility (cable) (125± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2

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QUALIFICATION INSPECTION (continued):	Requirements
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

	Type and Size	Feet (nominal)
	LSTSGU - 3	2500
	LSTSGU - 4 through 23	2000
	LSTSGU - 50	1500
*	LSTSGU - 75 through 150	1000
	LSTSGU - 200 through 400	800

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Agent:
DLA-IS

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